

Ejercicio. 8.1

$$\lim_{x \rightarrow 0} \frac{x}{\sin(x)}$$

$$\frac{a}{b} = \frac{1}{\frac{b}{a}} = \left(\frac{b}{a}\right)^{-1}$$

Solución.

$$\lim_{x \rightarrow 0} \frac{x}{\sin(x)} = \lim_{x \rightarrow 0} \frac{1}{\frac{\sin(x)}{x}} = \frac{1}{1} = 1$$

Ejemplo. 8.2

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{\sqrt[3]{x}}$$

Solución. $\sqrt[3]{x} = x^{1/3}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin(x)}{\sqrt[3]{x}} &= \lim_{x \rightarrow 0} \frac{\sin(x)}{\sqrt[3]{x}} \cdot \frac{x^{2/3}}{x^{2/3}} \\ &= \lim_{x \rightarrow 0} x^{2/3} \frac{\sin(x)}{x^{1/3+2/3}} \\ &= \lim_{x \rightarrow 0} x^{2/3} \frac{\sin(x)}{x} \\ &= \lim_{x \rightarrow 0} x^{2/3} \cdot \lim_{x \rightarrow 0} \frac{\sin x}{x} \\ &= 0^{2/3} \cdot 1 = 0 \cdot 1 = 0 \end{aligned}$$

Ejercicio 8.7

$$\lim_{x \rightarrow 0} \frac{1 - 2x^2 - 2\cos x + \cos^2 x}{x^2}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

Solución.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - 2x^2 - 2\cos x + \cos^2 x}{x^2} &= \lim_{x \rightarrow 0} \frac{1 - 2\cos x + \cos^2 x}{x^2} + \lim_{x \rightarrow 0} \left(-\frac{2x^2}{x^2} \right) \\ &= \lim_{x \rightarrow 0} \frac{(1 - \cos x)^2}{x^2} + \lim_{x \rightarrow 0} (-2) \\ &= \lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{x} \right)^2 - 2 \\ &= 0^2 - 2 = -2 \end{aligned}$$

Ejercicio 8.11

$$\lim_{x \rightarrow 0} x \cos x$$

Solución.

$$\lim_{x \rightarrow 0} x \cos x = \lim_{x \rightarrow 0} x \frac{\cos x}{\sin x} = \lim_{x \rightarrow 0} \frac{x}{\tan x} \quad \cos x = 1 \cdot \cos 0 = 1 \cdot 1 = 1.$$